

Technology and Healthcare

ENGR49580D Technology, Society and Sustainability

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- Drug Patents
- Orphan Drugs
- AI Diagnostics
- Electronic Medical Records
- Telemedicine
 - Robotic Surgery

Drug Patents

Rationale

- Like all other technologies; drugs, vaccines, and other medical technologies are patentable
- The nature of patents may vary and include, the chemical compound, the development or delivery method
- Drug patents are typically valid for 20 years, but may be extended
 - Existing drugs may be patented for new uses or new formulations
- Drug regulators (Health Canada / FDA) can protect drug patents by not licensing alternative products

Drug Patents

Rationale

- Like all other patents, drug patents are intended to incentivize innovation
- Having an assured period of exclusive benefit from an innovation is supposed to serve as an incentive for inventors to bring products to market
- Drug patents allow for the recovery of the high cost of getting regulatory approval for a drug

Drug Patents

Criticisms

- Drug patents contribute to the high costs of drugs
- ~80% of drug patent applications are not for new medicines
- Top selling drugs have multiple patents that extend protection to an average of 37.5 years
- Drugs costs drop by as much as 85% once generic version are available on the market

Drug Patents

Strategies for extending patents

- ‘Drip Feed’
 - New patent disclosures are filed with increased detail on previously disclosed technologies
- ‘Product Hoping’
 - New versions of the same product (eg higher dose, different delivery method, different release agent) are patented
 - Doctors are encouraged to prescribe the new versions
 - This prevents pharmacists for substituting similar (but not identical) generics

Orphan Drugs

- A classification for drugs used to treat rare diseases
 - Because of the rarity of these diseases the cost of these drugs is typically significantly higher since the development and approval costs are born by a smaller population
- Orphan drug programs make it more cost-effective to bring these drugs to market

Orphan Drugs

Definition

- US FDA: a drug that is “intended for the treatment, prevention or diagnosis of a rare disease or condition, which is one that affects less than 200,000 persons in the US” (~6 in 10 000 people)
- EU: intended for the diagnosis, prevention or treatment of a life-threatening or chronically and seriously debilitating condition affecting not more than 5 in 10,000 EU people

Orphan Drugs

Criticisms

- A course of treatment with an orphan drug can be as much as \$700 000 per year
- Thanks to government supports as well as a smaller trial population, this cost is not justified by the R&D, approval, or manufacturing costs

AI Diagnostics

- Artificial Intelligence and machine learning techniques are increasingly being used to analyze data from medical tests and imaging
- In some studies AI diagnostics are show to be more accurate that human doctors

In Class Activity

AI Diagnostics

- Would you trust AI diagnostics to assess test results for a critical illness in you or a loved one?
 - What about if this was a mature technology
- Do you believe that physicians will become depending on mature AI diagnostic technology the same way that we have become dependant on spellcheck?

Electronic Medical Records

- Most jurisdictions have migrated to some form of electronic medical record
 - These have mostly replace paper notes and records

Electronic Medical Records

Benefits

- Increased reliability
 - Electronic records are more easily searchable than paper ones
 - Electronic records can be backed up offsite to prevent loss
- Increased transferability
 - Electronic records are easily shared with specialists or transferred to new providers

Electronic Medical Records

Benefits

- Fraud prevention
 - Electronic records make it easier to detect duplication of billable services
 - Electronic prescriptions are harder to forge / alter
- Transparency
 - Electronic records can be easily made accessible to patients
 - Electronic records are more easily reviewed

Electronic Medical Records

Benefits

- Internal Security
 - Computer systems storing EMRs can identify unauthorized or unusual access
 - Computer logs may assist in tracking data breaches
- Encrypted storage may protect data
- Error detection algorithms can be employed to catch incorrectly entered data (unusual values) or copy-pasted data

Electronic Medical Records

Benefits

- Increased availability in emergencies
- Ability to identify contraindicated prescriptions / treatments

Electronic Medical Records

Criticism

- Electronic records are more vulnerable to unauthorized access
 - Loss / theft of connected devices
 - Hacking
 - Improperly wiped data on 'end of life' devices
- The use of 'out of jurisdiction' servers complicates regulatory compliance

Electronic Medical Records

Criticism

- Proprietary software may lock systems into a single vendor increasing costs
- Paper records are typically not converted due to cost and may become orphaned

Electronic Medical Records

Duty to Protect Privacy

- In most jurisdictions health information is protected by separate privacy legislation that other types of information
- In Ontario one of the following acts apply to non health information:
 - The Personal Information Protection and Electronic Documents Act
 - Freedom of Information and Protection of Privacy Act
 - Municipal Freedom of Information and Protection of Privacy Act
- Health information is governed by the
 - Personal Health Information Protection Act

In Class Activity

EMR and Electronic Privacy

- What type of electronic records are you most concerned about keeping private?
 - Social Media Activity
 - Online Dating
 - Chats (including photos) with an intimate partner
 - Financial Records (banking, credit cards, investments)
 - Medical Records (test results, diagnoses, prescriptions)

In Class Activity

EMR Monetization

- Would you be comfortable with healthcare provider monetizing your EMR by delivering targeted advertising? Similar to how Meta uses social media history and Google uses search history.

Telemedicine

- Refers to the use of technology to facilitate the interaction between a patient and a health-care provider who is not in the same room
- Applications vary and include:
 - Telephone consultations
 - Monitored at home diagnostic tools
 - such as blood pressure monitors and EKGs
 - Involved remote consultation suites with support staff

Telemedicine

Benefits

- Increased access to healthcare for at risk and underserved populations
 - Remote communities and First Nations
 - Prisons
- Increased safety in delivering healthcare
 - No risk of infectious disease transmission - pandemic management
 - In the case of prison medicine, reduced need to transport inmates

Telemedicine

Robotic Surgery

- Surgeon controlled robotic instruments are used to perform surgery
- Allows for smaller entry site and more precise work than can be done by human hands or other surgical techniques (eg Laparoscopy)
- Option to operate remotely to allow specialist to perform surgery with reduced travel need